DURATION: 20 MINUTES **FULL MARKS: 15(+2)**

ID:

1. You are given six CNN components and six scenarios. Match each scenario with the single component that best addresses it and briefly justify your choice. Use each component in exactly one scenario and match each scenario with exactly one component. 9

Aggressive Stem	Multi-Scale Branches
Global Average Pooling	Bottleneck Blocks
Auxiliary Classifiers	Residual Blocks

1. You see that your model's accuracy has saturated at its current depth, but your GPU cannot accommodate any additional parameters.
2. You see from profiling that most of the computation is spent in the first few high-resolution layers.
3. You see that a 56-layer plain CNN has higher training error than its 20-layer version.
4. You see that most of the model's weights are concentrated in the classifier head rather than the convolutional backbone.
5. You observe that layers far from the output end are learning very slowly.
6. You see that a detector handles small details well but often misses larger visual patterns, and vice versa.

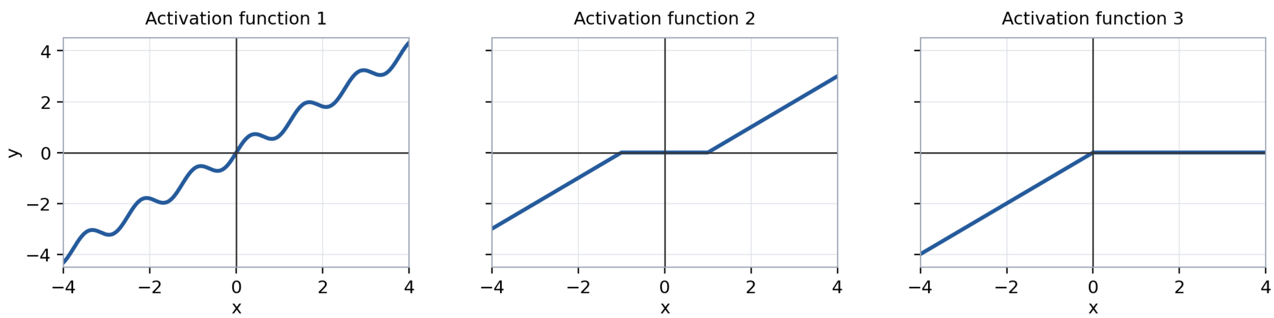
Solution:

Expected Answer:

Scenario	Feature
1	Bottleneck Block
2	Aggressive Stem
3	Residual Blocks
4	Global Average Pooling
5	Auxiliary Classifiers / Residual Blocks
6	Multi-Scale Branches

Marking Rubric (9 marks): 1.5 marks for each correct feature selection and justification.

2. **Question B.** The three activation functions below are used in three neural networks. For each of these activation functions, write any potential issues the neural network may face, if any. 6



Solution:

Expected Answer:

- **Activation function 1:** The oscillation makes the derivative change sign and creates non-monotonic regions, so gradient directions can reverse locally. Its derivative can also become large or small depending on x.
- **Activation function 2:** The flat region around zero has zero derivative, so inputs in that interval receive no gradient. The joins at $x = -1$ and $x = 1$ are also non-differentiable.
- **Activation function 3:** The function has zero derivative for all positive inputs, so units receiving positive inputs cannot learn through this activation. It is also non-differentiable at $x = 0$. Although this is just ReLU flipped, so this will work equivalently well as ReLU and is considerably better than the other two.

Marking Rubric (6 marks): 2 marks for each graph's issue, including a valid explanation of the underlying gradient or optimization problem.

3. **(Bonus Question)** Write the two worst features of this course (*CSE 4835: Pattern Recognition*). 2

Solution:

Expected Answers:

1. No ice cream.
2. No ice cream.